Compliance sheet — star-mat3

Everything a supplier qualification, a legal review or an export check usually asks about star-mat3 0.1.0, in one document.

1. Identity

Document	STAR-PKG-STAR-MAT3-EN · v1.2 · 2026-10-15
Package	star-mat3
Version	0.1.0
Capability	star.geometry.mat3
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_mat3

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE and NOTICE files, keep the copyright notices, and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import
hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_mat3-0.1.0-
py3-none-any.whl
```

FILE	SHA-256
star_mat3-0.1.0-py3-none-any.whl	d5a0688a0f8aa4960a8fe4647e683c8e55085b12c7339671253e20aebd2fdd31
star_mat3.py	40456aa4d4da89cd6b1aaeca81e93b57e971b8cf672d9b34c005a5eb5b5ee343
star_mat3-0.1.0-source.zip	5e4c4cd46e0cac7000da0ec164631aa8fb6a8285fad091cd7622ffe08feade27

5. What it does, and its limits

Transpose, exact determinant and inverse, products and a proper-rotation test for 3×3 matrices; determinant and inverse identical to exact rational arithmetic on 575 matrices including condition numbers up to 1e8 (where SPICE's inverse is off by 6e-6 and NumPy's by 9e-10); singular matrices refused.

3×3 only; no eigenvalues, no decompositions, no orthonormalisation of a drifting rotation. The exact arithmetic makes `inverse` slower than a floating-point one (tens of microseconds): it is for correctness, not for inner loops. "Exact" means the exact inverse of the floats given: if those floats are themselves rounded measurements of an ill-conditioned matrix, the inverse amplifies their error by the condition number, and nothing here can prevent that. The products are compensated sums of rounded products, accurate to a few units in the last place, not exact.

- `determinant` and `inverse` in exact rational arithmetic (a float is an exact fraction), rounded once: correct to the last digit whatever the conditioning. `inverse` refuses singular and nearly singular matrices, with a test that does not depend on the scale of the entries.
- `transpose`, `matmul`, `matvec`, `identity`; `is_rotation(m, tol)` is True only for an orthonormal matrix with determinant +1 within the tolerance.
- Measured on 575 matrices (300 general, 150 rotations, 125 with condition numbers up to 1e8) against EXACT arithmetic (SymPy): determinant and inverse identical to the exact value rounded; products within 4e-15. On the ill-conditioned matrices SPICE's inverse differs from the exact one by up to 6e-6 of its largest entry and NumPy's by 9e-10. `is_rotation` agrees with SPICE on every matrix.

- Every function returns floats (or a bool) or raises `ValueError`: wrong shape or type, an entry that is not a finite real number within ±1e60 (booleans refused), a singular or nearly singular matrix, an inverse that overflows, a tolerance outside (0, 1].

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_mat3_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-mat3}: The 3×3 matrix operations of frame and attitude code},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_mat3/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

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